

QUESTION 1 [17 marks]

- a. **(8 marks)** The circuit in Figure 1 represents the DC equivalent of a simple residential power distribution system which allows the operation of appliances operating at 120V or 240V. Appliances are represented as resistors. Calculate the current through each of the appliances (i_{lamp} , i_{heater} , i_{oven}).

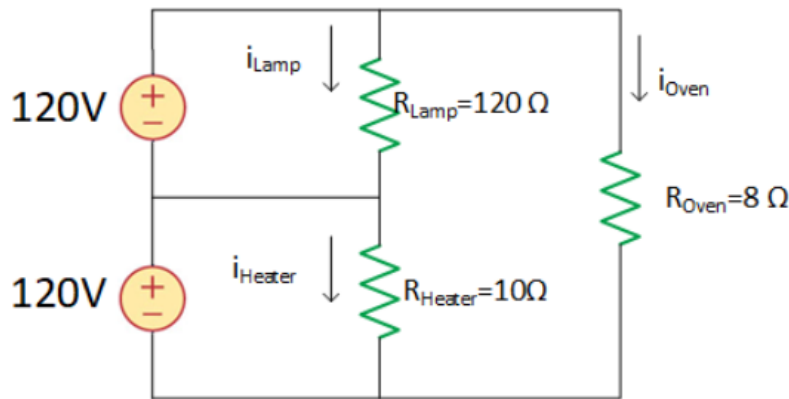


Figure 1

- b. **(9 marks)** The circuit shown in Figure 2 is a battery-operated LED circuit for a torch. The torch is powered by a 6-V battery and has white LEDs with forward voltages across their terminals equal to 1.7 V. The maximum current that can flow through the LEDs is 20mA.

Calculate the resistor value R that will ensure maximum brightness of the LEDs, where brightness is proportional to current flow (i.e., the higher the current, the brighter the LEDs).

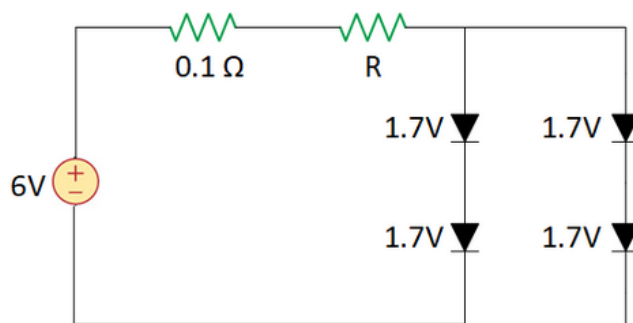


Figure 2

QUESTION 2 [20 marks]

- a. **(10 marks)** For the circuit shown in Figure 3, apply nodal analysis and write down the nodal equations using the labels provided.

Note: Simplify the equations in the form $av_1 + bv_2 + cv_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but **DO NOT** solve them.

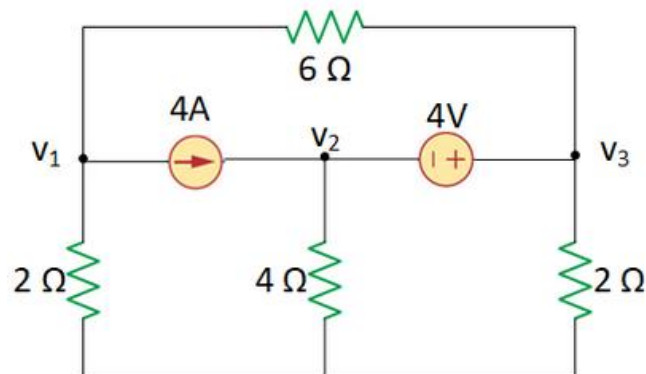


Figure 3

- b. **(10 marks)** For the circuit shown in Figure 4, apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but **DO NOT** solve them.

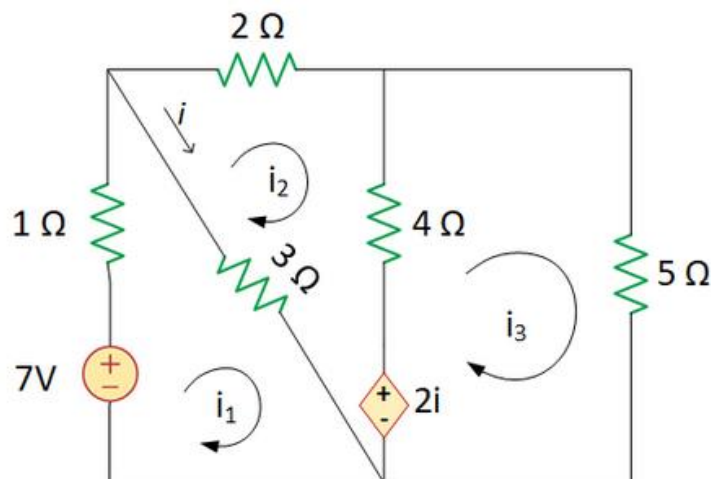


Figure 4

QUESTION 3 [18 marks]

The circuit in Figure 5 corresponds to a bicycle with front and rear bulb lights and a permanent magnet generator (hub dynamo) to power them. The circuit also includes a USB charger to charge the mobile phone while riding.

When the bicycle is travelling at 20km/h (average cycling speed), the permanent magnet generator voltage is 5 V.

- (16 marks)** Use superposition to calculate the current flowing through the front bulb, i_{front} .
- (2 marks)** Calculate the power dissipated by the front bulb (i.e., 20Ω resistor).

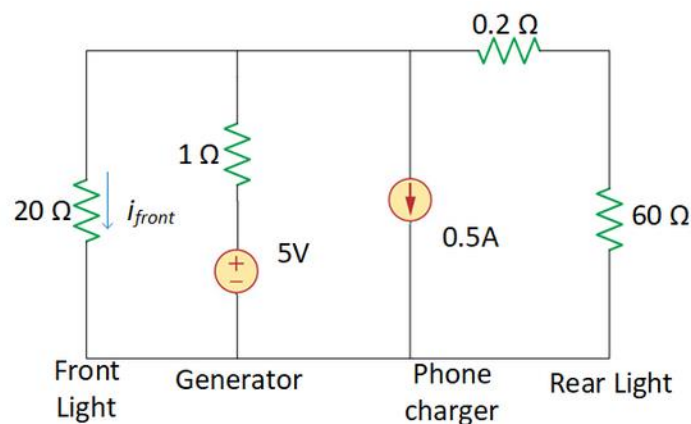


Figure 5

QUESTION 4 [18 marks]

The circuit shown in Figure 6 is being used to power a water heater. The water heater can be modelled as a single resistor connected to terminals a - b .

- (8 marks)** Calculate the Thevenin voltage at terminals a - b .
- (8 marks)** Calculate the water heater resistance that will ensure the maximum transfer of power from the circuit to the heater.
- (2 marks)** Find the maximum power that can be delivered to the heater.

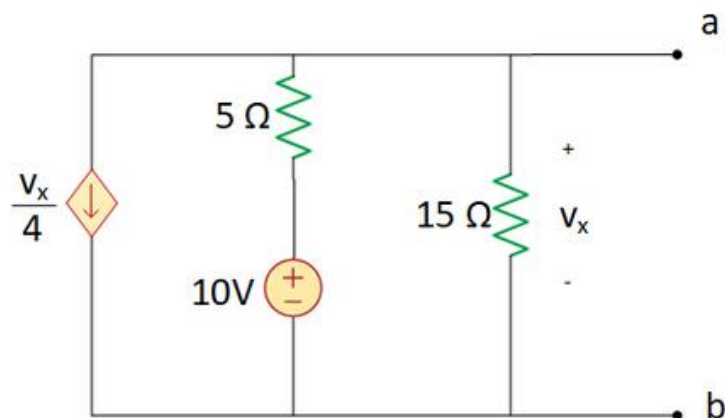


Figure 6

QUESTION 5 [27 marks]

- a. **(12 marks)** The switch in the circuit shown in Figure 7 has been in position *a* for a long time. At $t = 0$, the switch moves instantaneously to position *b*. Find $v_o(t)$ for $t > 0$.

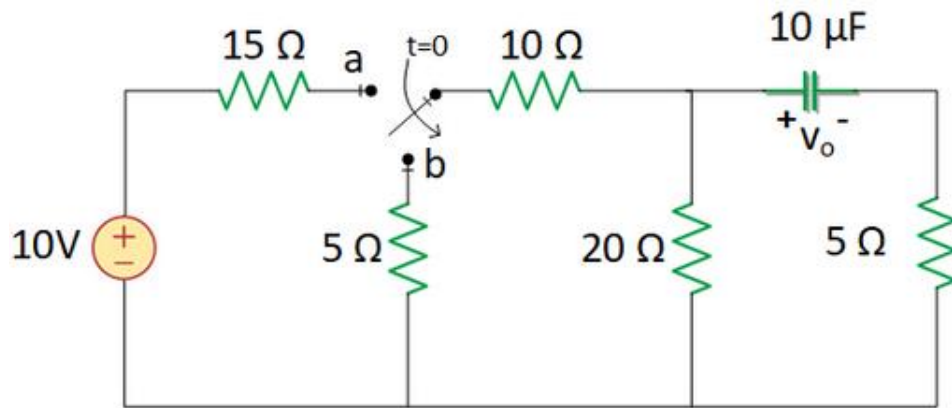


Figure 7

- b. **(15 marks)** The RC circuit in Figure 8 represents the intermittent windshield wipers of a car. The output voltage of the capacitor, v_{out} , is used to control a switch. The switch is normally open, but when the output voltage of the capacitor reaches 10 V, the switch closes, energizing an electric motor and discharging the capacitor. The motor causes the wipers to sweep across the windshield. The wipers sweep for 10 seconds, and after that the switch opens and the capacitor begins to charge again.
- i. **(12 marks)** If the capacitor is initially discharged, how long will it take for the switch to close (i.e., how long will it take for the capacitor to reach 10V)?
- ii. **(3 marks)** Once the switch closes, calculate the time constant of the RC circuit.

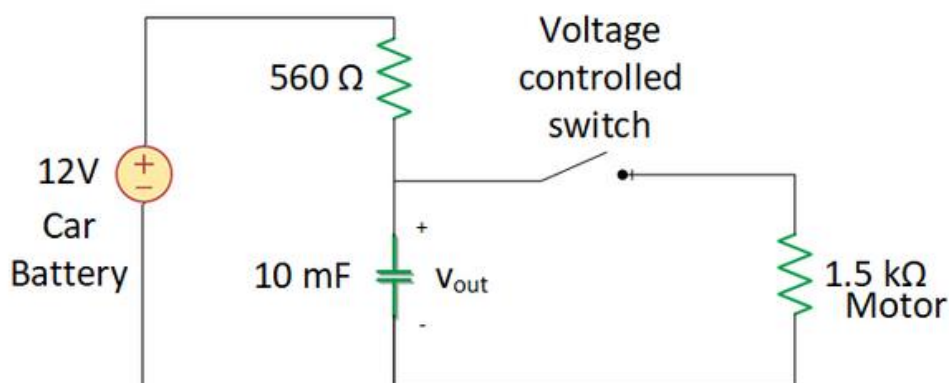


Figure 8